

Huiuk Jang

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EDUCATION

DigiPen Institute of Technology, Redmond, WA, US & Keimyung University Daegu, Republic of Korea

B.S. in Computer Science in Real-Time Interactive Simulation

April 2027

Relevant Courses: Advanced C/C++, Data Structures, Computer Graphics, Network, Calculus, Linear Algebra, Physics

TECHNICAL SKILLS

Programming Languages and Libraries: OpenGL, C++, C, Python, C#, raylib

Engines: Unreal Engine 5.4, 5.7, Unity 2019.4, Unity 6.3(With Nintendo Class), Unity 2022.3.47f1, Custom OpenGL Engine

WORK EXPERIENCE

Programming Intern | Tactical Affairs | Redmond, WA

May - Jul 2026

Project Escape | Unreal Engine 5

- Engineered core gameplay systems in Unreal Engine using blueprints, including shield/HP status managers, player rail-movement mechanics, HP pickups, and mini maps.
- Authored custom shaders for metallic visual effects and configured lighting setups to establish art direction and improve visual fidelity.
- Resolved critical GPU bottlenecks and stability issues in Unreal Engine, eliminating heavy overdraw to drop scene complexity from Very High to Normal and debugging GPU crashes for seamless level resets.
- Designed interactive level layouts and game assets, optimizing spatial design for balanced player progression.

Fire and Ice | Unity

- Configured and integrated shield mechanics for enemy units based on designer specifications, including randomization, adjusting attributes to balance combat encounters, in Unity using C#.
- Validated gameplay balance and movement parameters by reviewing player progression data with designers, providing feedback to refine game pacing.

Operation Staff | DigiPen Institute of Technology | Redmond, WA

Jun - Aug 2026

- Managed lunchtime operations for a 9-week summer camp, ensuring a safe, hygienic, and organized environment for 40+ campers in collaboration with team members.

Teaching Assistant | DigiPen Institute of Technology | Redmond, WA

Sep – Dec 2025, Jan - Apr 2026

- Mentored students in C++ to develop independent debugging and high-level programming skills.
- Addressing student questions effectively and efficiently, and inducing students to solve python problems.
- Grade students' homework in Mathematics in Music & Sound class.

PROJECTS EXPERIENCE

Programmer | Wing Dragon Heist | 3D Platformer

Sep 2025 - April 2026

- Integrated game assets and developed custom Cel-shaded materials in Unreal Engine using blueprints to achieve a stylized visual aesthetic.
- Architected diverse particle systems using Niagara for player actions, hit effects, projectile effects (arrows), climbable walls, and dynamic UI elements (HP/Stamina bars).
- Developed UI for Key items and text systems that trigger based on player inventory and environment state.
- Modified environment shaders and gameplay systems, including enemy spawners and light-based mechanics that prevent "Shadow Mode" entry.

Programmer & Engine Developer | ABRAXAS | 2D Platformer Roguelike

Sep 2024 - Jun 2025

- Designed and implemented graphic rendering, audio processing, data input systems, camera control, game windows, and inputs using OpenGL for a custom game engine.
- Developed a character selection screen interface, game numerical adjustment objects, and icons.
- Debugged and Resolved specific object collision issues-prevented all other objects from passing through this object.

Programmer& Engine Developer | Shadow Dimension | 2D Platformer Puzzle

Mar - Jun 2024

- Contributed to the creation of a modular game engine and core player-object interaction system using raylib library and C++.
- Developed a 3D interaction detection and resolution system and visualization system for top/side view properties in 3D Modulus.
- Created a dynamic data update system that allows real-time changes using external files.

Programmer | Mine Mine | 2D Platformer Arcade

Sep - Dec 2023

- Engineered core 2D gameplay systems in C++ using Raylib, including player movement, object interaction mechanics, and custom 2D collision detection and resolution.
- Developed graphics and level mechanics, implementing procedurally generated tile maps, conditional image conversion, dynamic light diffusion, and distance based visibility limitations.
- Built a developer debug mode featuring real-time player speed adjustment, game time control, and instant object interaction tools to accelerate testing and debugging.